



Ecological finance theory: New foundations

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ABSTRACT

This paper puts forth *Ecological Finance Theory* as a new vision for 21st century financial economics. We first motivate this project by referring to the current global bio-geophysical context. We then introduce new meta-theoretical hypotheses, which we compare with those of the previous paradigm. We then borrow concepts of resilience, diversity, self-thinning, self-regulated mitosis, and ecological transparency from biology and introduce them in the field of financial economics. Finally, we analyze the impact of an increase in ecological transparency on resilience and diversity by developing a new stock-flow consistent model with mitotic control. The present paper is hence an invitation to reorient financial theory in line with the Anthropocene context.

1. Introduction

In recent decades, the *Great Acceleration* of socioeconomic trends has caused a set of ‘regime shifts’ in biogeochemical and social systems, creating a cascade effect and causing an abrupt deterioration in human welfare¹ (Lenton and Latour, 2018, Rockström, Steffen, Noone, et al., 2009, Steffen et al., 2005). Rescaling human actions and behavior to the load capacity of the biosphere, from the local to global levels of every scale is now widely recognized as a major civilizational stake (Clark and Clark & Munn, 1986, Folke et al., 2010, Steffen et al., 2011, Galaz et al., 2015).

Earth scientists have identified financialization (Krippner, 2005) as one of the key mechanisms by which human activity undermines the biosphere (Nyström et al., 2019). Multilateral institutions have also unanimously acknowledged the need for a structural reform of the global financial system.² At the time of writing, finance research has

responded mainly by analyzing the effects of Environmental, Social and Governance (ESG) criteria on shareholder wealth (Friede et al., 2015).

This strand of research has included a new dimension (‘sustainability’) to the array of investor decision criteria (along with price, expected rate of return, portfolio diversification, equilibrium return, risk premium). Recent studies have documented deviations in the price and returns of ‘green’ financial instrument from standard financial instruments (Baker et al., 2018; Karpf and Mandel, 2017), identified new equilibrium return for shareholders using asset pricing formula (Zerbib, 2020), or analyzed the evolving preference of shareholder for sustainability under various governance frameworks (Brière et al., 2019).

A budding consensus, however, argues that the climate crisis calls for a much deeper reorientation of finance research. Given the complexity and unforeseeable nature of feedbacks occurring within the Earth System, identifying sustainability with market brands and labels appears self-defeating. Indeed, current accounting systems do not identify what is

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¹ A recent survey by Future Earth (2020) of more than 200 international scientists has identified five global risks that have the potential to impact and amplify one another, in ways that might cascade to create global systemic crisis. These risks include failure of climate change mitigation and adaptation; extreme weather events; major habitat and biodiversity loss and ecosystem collapse; food crises; and water crises.

² For instance, the European Commission High-Level Expert Group on Sustainable Finance (HLEG) famously underlined that “reaching our Paris agreement goals requires no less than a transformation of the entire financial system, its culture and its incentives” (HLEG, 2018). The UNEP-Finance initiative, which aims to foster Principles for Responsible Banking (PRB), Principles for Sustainable Insurance (PSI), and Principles for Responsible Investment (PRI) in order to interlock financial systems with the objectives of the 2030 Agenda for Sustainable Development, is another example. Efforts to reform the financial system are part of a broader transformation agenda, framed around the sustainable development goal agenda (SDGs), which was adopted by all 193 UN Member States on September 25, 2015. The SDG agenda provides a new narrative for economic policy, organized around six major societal transformations, in the fields of education, health, energy systems, agriculture, urban planning and technology - all supported by good governance, peace and international cooperation (SDSN, 2018). These transformations are measured via 17 goals, which are disaggregated into a global indicator framework currently including 231 indicators. This framework was agreed upon at the 48th session of the United Nations Statistical Commission held in March 2017, and reviewed in March 2020. (United Nations Statistical Commission, 2020).

'sustainable' – let alone assign it a monetized value (Nyström et al., 2019). Rather than superimposing a set of contingent financial norms and criteria on to the analysis of the Anthropocene (i.e. defining sustainability as an independent variable and leaving financial metrics as a dependent variable), one may need instead to *turn the order upside down*. This implies to set desirable financial and economic system features as independent variables, and observed climatic, economic and social trends as dependent variables, in the search for a new resilient model.

Mainstream academic finance is thus in a state of *paradigmatic crisis*.³ However, “to reject one paradigm without simultaneously substituting another is to reject science itself.” (Kuhn, 1962). This paper thus proposes *Ecological Finance Theory* as a new theoretical framework.⁴

To do so, we apply the analytical structure of Burrell and Morgan (1980) and Ardalan (2008) and introduce new meta-theoretical framework, concepts, methodologies, and research questions in finance studies. We first document the meta-theoretical shortcomings of neo-classical finance theory and introduce a set of alternative pre-suppositions. We then import a set of biological concepts in the field of financial economics. Finally, we show how this approach can help reorient finance research through a simple modelling exercise.

The present paper is an invitation to the reader to adopt these new hypotheses, further analyze the implications of the proposed concepts (or propose new ones), and hence contribute to the development of *Ecological Finance Theory* as a new paradigm in finance research. The remainder of the article is structured as follows. The second section defines a paradigm, describes the main features of the neoclassical paradigm, and explains why it cannot tackle climate issues. The third section introduces new meta-theoretical hypotheses for *Ecological Finance Theory*. The fourth section borrows concepts of resilience, self-thinning, mitosis and transparency from biological meanings, and introduces them in finance research. The fifth section articulates these new notions by building a simple ecological finance stock-flow consistent model (*'Philia 0.1'*). The sixth section brings together our conclusions and discusses future development prospects for *Ecological Finance Theory*.

³ This paradigmatic failure may not be visible due to the pervasive influence of 'leading' finance journals which do not reflect the diversity of finance scholarship (Hu et al., 2021). According to its critics, the modern finance paradigm is dysfunctional due to (i) the *logical positivist bias* (i.e. blindness to the entanglement of facts and values), (ii) the *methodological individualism fallacy* (i.e. failing to account for top-down causation), (iii) the *shareholder value maximization dogma* (i.e. failing to acknowledge the diversity of economic organizations and rationalities), and (iv) loyalty to the *efficient market hypothesis* (in spite of its falsification by several studies). For discussion in books, see, for instance, Bourghelle et al. (2021), Chambost, Lenglet, and Tadjeddine (2019), Alijani and Karyotis (2016), Paraque and Pérez (2016), Dembinski (2009), MacKenzie (2006). For discussion in articles published in finance journals, see for instance, Lagoarde-Segot (2015, 2019); Lagoarde-Segot and Paraque (2018); Revelli (2016); Walter (2016, 2019); Dupré and Perluss, (2016). For discussion in articles published in ecological economics and sustainability journals, see for instance Svartzman et al. (2019); Lagoarde-Segot (2020); Chenet et al. (2021).

⁴ *Ecological Finance Theory* is the outcome of several years of friendly, formal and informal discussions within the *Post-Crisis Finance Network*. In 2015, this group of researchers (based in France, but connected to various international networks) issued a Manifesto identifying the meta-hypotheses of neo-classical finance as problem in the emerging global ecological, social and economic context. This group has then gathered in several meetings, held under the auspices of various institutions and associations, such as *SDSN France*, the French *Fondation Nationale pour l'Enseignement de la Gestion des Entreprises*, the *Association pour Renouveler la Recherche et l'Enseignement en Economie et en Finance* (AREF) at the University of Fribourg, the *Responsible Investment Forum*, the AFEP/IIPPE conference, the RIODD conference, and successive INFINITI conferences in Aix-en-Provence, Prato, Ljubljana, Dublin and Valencia. The development of *Ecological Finance Theory* has also greatly benefited from informal interactions with students and colleagues.

2. The existing paradigm

2.1. Definition of a paradigm

Scientific paradigms can be defined and characterized as “a set of fundamental assumptions, which, in turn, translate into certain rules and standards for scientific practice” (Ardalan, 2008, p.31). Within a paradigm, researchers employ a specific jargon to formulate a set of acceptable set of research hypotheses, and agree on the procedures by which to test these hypotheses. A paradigm thus articulates three basic elements: *meta-theoretical hypotheses*, *metaphors*, and *puzzle-solving activities* (Ardalan, 2008; Burrell & Morgan, 1979; Morgan, 1980):

- *Meta-theoretical hypotheses* refer to a shared and tacit world vision governing the norms and rules of scientific research at a very fundamental level. This world vision is related to a set of tacit hypotheses, regarding the very essence of the phenomena under investigation (*ontology*); and the definition and the role attributed to knowledge (*epistemology*);
- *Metaphors* (or concepts) build upon these meta-theoretical hypotheses to provide known mental images for understanding or for studying the research object.⁵ Metaphors structure scientific activities by orienting research towards the analysis of similarities (and discrepancies) between the object of study and the shared jargon;
- *Puzzle-solving* consists in the examination, operationalization and measurement of detailed implications of the metaphorical insight upon which research is based. Puzzle-solving debates concern the identification and proper implementation of the appropriate methodological tools required to examine a particular research question.

Scientific paradigms have a nested structure: controversies around puzzles are framed in a shared jargon; and the latter stems from meta-theoretical hypotheses. In normal periods, these meta-theoretical hypotheses do not need re-examination. This was the case during the last decades, where the modern (or neoclassical) paradigm was the dominant paradigm in finance research. During this phase, research focused on applied and methodological topics, occasionally introducing new metaphors to describe their object of study (such as those of behavioral finance).

2.2. Features of the existing paradigm

The neoclassical paradigm analyzes financial data in terms of causes and effects, making knowledge-claims based on quantitative methods derived from the natural sciences. Its three key meta-theoretical hypotheses include *objectivism*, *reductionism* and *instrumentalism*. *Objectivism* is the notion that financial markets exist as natural and external entities, independently of the representations that individuals have of it. *Reductionism* is the belief that the whole is no more than the sum of its parts. It assumes that large entities (such as society or the Earth system) have no real substance, which rules out any possible top-down causation. Finally, *instrumentalism* is the notion that assumptions based on “wildly inaccurate description of reality” (Friedman, 1953, p.14) are

⁵ We could say that the “greenhouse” effect is a metaphor often used to explain climate change dynamics. “Cascade effects” is another metaphor to visualise a linked series of cause-effect process.

sound so long as they permit to calculate an equilibrium position.⁶

These meta-hypotheses are intertwined with a specific worldview presenting the financial system as a market for loanable funds, where banks play the role of intermediaries. Unless thwarted by 'rigidities', it is believed that the financial system discovers the true model of the economy, and reveals it to all in the form of a vector of prices. Actors (banks and markets) hence process all relevant information generated by the socioeconomic and ecological sub-systems, and translates it into a system of 'true' prices through some kind of aggregate investor utility function (Fig. 1). Short-run deviations from the natural state of equilibrium may occasionally occur; but these are seen as resulting from stochastic, exogenous shocks (e.g. metaphorical 'black swans').

This worldview is informed by the following keystone contributions:

- (i) The financial system is a market for loanable funds (Walras, 1874);
- (ii) The natural state of markets is general equilibrium (Arrow-Debreu, 1954);
- (iii) Financial markets eliminate risk through diversification (Markowitz, 1952);
- (iv) There is market equilibrium level for investor profits (Sharpe, 1964; Fama and French, 1996);
- (v) Financial markets price reflect fundamental value (Fama, 1970, 1990);
- (vi) Social welfare equates shareholder value (Jensen and Meckling, 1976; Jensen, 2001),
- (vii) Risk is a Brownian process (Black, Scholes and Merton, 1973)

Based on this system of beliefs, the neoclassical finance paradigm expresses research hypotheses using specific metaphors, such as 'financial repression', 'agency problem', 'signaling', 'efficiency', 'deviation', 'equilibrium', 'rational bubble', 'black swan', 'performance' and, more recently, 'sustainability'. Sustainable finance research hence consists in documenting how a variety of 'sustainability metrics' and evolutions of 'investor preference' affect the 'price discovery' process, leading up to 'deviations' from the 'equilibrium return', or to a change in the equilibrium itself (Friede et al., 2015). The implicit assumption is that financial markets, by virtue of their assumed properties would naturally uncover and reveal the monetary value of the Earth System.

2.3. Sustainability and the neoclassical paradigm

The state of the world stands in sharp contrast with the above-described system of beliefs. A simple linear regression between global financial development (broad money to GDP ratio) and CO2 emissions kilotons has a correlation coefficient of 0.953 (Fig. 2). A linear regression between global capitalization of domestic firms and CO2 has a correlation coefficient of 0.960 (Fig. 3). Economic growth also affects CO2 emissions. The correlation coefficient between world GDP and CO2 emissions is 0.9667 (Fig. 4). We also know that CO2 emissions mirror other indicators of earth system destabilization (such as nitrous dioxide

emissions, ozone depletion, territorial biosphere degradation, tropical forest loss, ocean acidification) (Steffen et al., 2005).⁷ This destabilization has come to threaten the reproduction of life on Earth.⁸ The strength of the observed correlations indicates that financial markets have failed to incorporate the threat of systemic bio-geophysical risk.

A voluminous literature has discussed how GDP growth and financial development have triggered the destabilization of the Earth System. First, the global economy is a worldwide social-ecological network of actors and institutions spanning sectoral, jurisdictional and geographical boundaries known as the Global Production Ecosystem (GPE). In recent years, the Global Production Ecosystem has responded to increased demand for biomass by intensifying other inputs used by humans (such as sand, carbonates, water, fossil fuels, fertilizers, pesticides, antibiotics and technology); thereby decoupling from the natural processes sustaining it (Nyström et al., 2019).

Second, the GPE is steered by a set of vertically and horizontally integrated transnational corporations. Many of these corporations are financial: as a collective, the "Financial Giants", through their common block holding power, have a considerable influence on companies shaping biomes critical for the stability of the climate system (such as the Amazon forest in Brazil and the boreal forests in Russia and Canada) (Galaz et al., 2018). The governance of the GPE and the legal framework within which it operates are tightly connected to the neoclassical finance paradigm (for instance through 'shareholder value governance').

Third, the acceleration of financial market development (a phenomenon known as 'financialization'⁹) has also contributed to decoupling. For instance, agricultural derivatives abstract food from its physical form into highly complex commodity derivatives, whose prices variation affect the global food system (Russi, 2013). Other research has indicated that the rise of fintech (for instance through high-frequency trading) induces a 'disjuncture' between the representation of value and the actual production of value by biological processes (Lagoarde-Segot and Currie, 2018).

One may of course wonder why finance research has failed to consider this tremendous 'market failure' (Stern Review, 2007). One explanation lies with ideology: due to in part to their own conservative leanings, and in part to ever-growing pressure of journal rankings which reinforce the influence of leading (neoclassical) finance journals, many finance researchers still accept that financial markets yield efficient outcomes, even if this requires absurd assumptions.

This is the old 'joint hypothesis' problem: any test of 'market efficiency' involves an asset-pricing model (Fama, 1991). However, as

⁷ The *Anthropocene* (the notion that the activities of humankind have given age to a new geological age) was initially proposed by Nobel laureate Paul Crutzen and officially recognized by the International Geological Congress in Cape Town in August 2016 (Maitland, 2019)

⁸ The International Geosphere Biosphere Program indicates that "the Earth System has recently moved well outside the range of the natural variability exhibited over at least the last half million years" (IGBP, 2004, p.3) The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) reports that the rate of extinction is currently tens to hundreds of times higher than the average over the last 10 million years. About 1 million species face possible extinction within decades because of deforestation, direct human exploitation of organisms, climate change, pollution, and the invasion of ecosystems by alien species due to global transport (IPBES, 2019). Between 2030 and 2050, climate change is expected to cause approximately 250,000 additional deaths per year, from malnutrition, malaria, diarrhea and heat stress alone (World Health Organization, 2018).

⁹ Financialization refers to "the increasing importance of financial markets, motives, institutions and elites in the operation of the economy and its governing institutions" (Epstein, 2005), to a "pattern of accumulation in which profits accrue primarily through financial channels than through trade and commodity production" (Krippner, 2005), or, more generally, to "the increasing dominance of financial actors, markets, practices, measurements and narratives at various scales, resulting in a structural transformation of economies, firms (including financial institutions), states and households" (Aalbers, 2015).

⁶ Examples of unrealistic assumptions used in neoclassical finance models include *atomicity* (the market comprises a multitude of investors who, individually, have no influence of the asset price discovery process); *perfect information* (all participants on the financial market have, at all times, access to all the relevant information on the evaluation of tradable assets); *absence of transaction costs* (at all times, investors are at liberty to enter and exit the market at no cost to them); *homogeneity* (the various assets traded have similar institutional and fiscal characteristics); *rationality* (investors conduct themselves in a rational, self-serving manner); *certainty* (the risk of debtors defaulting is assumed to be zero).

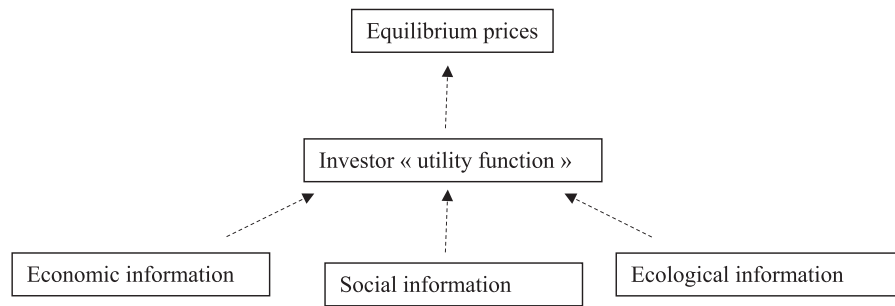


Fig. 1. The neoclassical finance outlook.

Note: this figure shows the interlocking of the socioeconomic and environmental system within the financial system as implied by neoclassical finance theory.

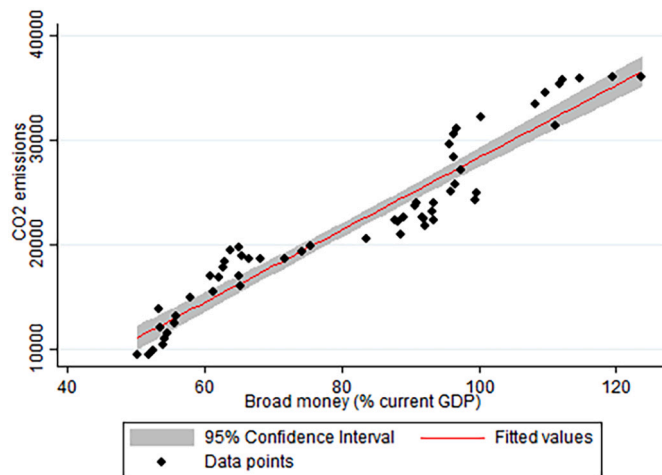


Fig. 2. Atmospheric CO2 emissions and financial development 1960–2019 (correlation 0.953, p -value 0.000).

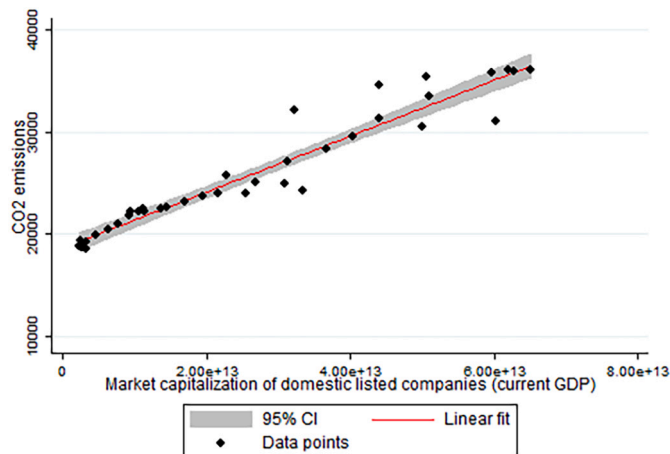


Fig. 3. Atmospheric CO2 emissions and market capitalization 1980–2019 (correlation 0.960, p value 0.000).

highlighted in Tobin (1971), these models (such as the CAPM) assume homogenous investor preferences, and have hence zero trade in equilibrium. This assumption – which results from the ‘representative agent’ fashion in 1970’s–1980’s macroeconomics – is at odds with the daily functioning of financial markets, where traders take or unwind positions based on conflicting information and expectations.

Another simpler – and perhaps more controversial – explanation has to do with money. In the words of former speculator George Soros

(2010, p.86) “by far the more powerful force working in favor of market fundamentalism is that it serves the self-interests of owners and managers of capital”.

3. Ecological finance theory

3.1. Meta-theoretical hypotheses

3.1.1. Ontology

Neoclassical finance theory builds up aggregate relationship from a set of expected microeconomic relationships. These relationships typically stem from assumed ‘constrained optimization’ of a ‘representative agent’. Ecological Finance Theory, on the other hand, rejects both instrumentalism and reductionism and begins with the following Earth Science observations (Fig. 5):

- The Earth System is the *higher order-system*,¹⁰ containing all the other systems (including the atmosphere, geosphere, hydrosphere, the biosphere, as well as their complex interactions). The Earth system is more complex, works at slower rates, is less flexible, and displays less substitutability than the other sub-systems. For instance, some forms of biodiversity are local, some are global, and the complex relationship between local and global biodiversity declines is not yet fully understood (Cardinale et al., 2018);
- The production system is a sub-system of the biosphere in which the human species derives a flow of the energy from the Earth System, throwing back waste and other emissions. Once condensed as matter, this flow of energy circulates in the form of labor, goods and services, which are exchanged for money.¹¹ The production system includes a wide range of activities, including market and non-market, paid and unpaid activities, undertaken by human-agents and organizations of all sizes, within the context of prevailing social norms, rules, culture and law;

¹⁰ In cybernetics, any system consists in a set of inter-related subsystems, each subsystem being also hierarchic in structure. The whole systemic structure can thus be decomposed until one reaches some lowest level of elementary sub-systems (Simon, 1962). One can identify higher-level systems based on properties such as higher complexity and richness of information. Living systems (from an anthropocentric point of view) are interlocked in the following hierarchical order: the biosphere, the socioeconomic sphere, smaller social units, and finally, the individual human being. Managing a higher-level system with the signals emitted by a lower-level sub-system leads to a loss of consistency and to the destruction of the higher-level system, due to a loss of information (Dron, 2015).

¹¹ For instance, the energy potential of biofuels dissipates in the form of the heat produced when fuels combine with atmospheric oxygen, allowing the different phases of the economic cycle to take place (energy industry, production of intermediary goods, and production of final goods, foods (fuel of living organisms), consumption, and wastes) (Bardi, Falsini, & Perissi, 2019).

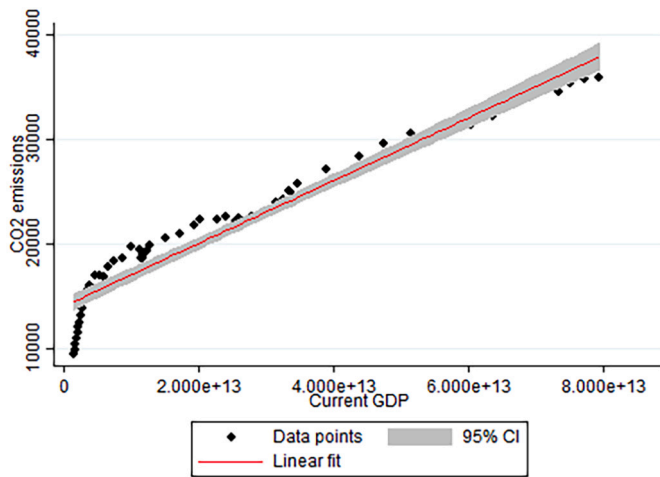


Fig. 4. Atmospheric CO₂ emissions and global GDP 1960–2019 (correlation 0.966, pvalue 0.000).

Source: Authors' calculation based on World Bank data (data.worldbank.org).

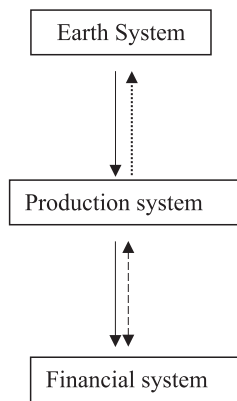


Fig. 5. A new ecological ontology for finance theory.

Note: a full line indicates a flow of energy derived from the biosphere. A dotted line indicates a flow of human waste and emissions. A dashed line indicates a flow of money.

- The financial system comprises a legal and accounting system, and institutions such as banks and financial markets. It records the monetary price of the actual (or expected) value of various items circulating in the production system. It also issues, circulates, and destroys the money tokens necessary for these transactions. While the organization of our contemporary production systems is *finance driven*; the financial system is nonetheless biologically dependent upon the other systems.¹²

3.1.2. A new worldview

Table 1 highlights presuppositions differences between Ecological Finance Theory and Neoclassical Finance Theory on a variety of topics such as markets, money and banks, governance, modelling, ethics and the micro/macro link. The second column illustrates the neoclassical position by quoting major neoclassical finance authors (such as William Sharpe, Eugene Fama, Milton Friedman, Michael Jensen, William Meckling, John Gurley, and Edward Shaw). The third column substantiates the ecological finance position on these topics with references

such as economics 'Nobel Prize' winners Elinor Oström and Amartya Sen, mathematician (Nicholas Georgescu-Roegen), climate scientists (Magnus Nyström et.al), ecologists (Simon Levin et.al), and monetary supervision institutions (the Bank of England). Inspection of Table 1 highlights that neoclassical finance theory builds on the loanable fund theory and assumes that financial markets spontaneously find their equilibrium (unless thrown off by a shock external to their own logic). It sees shareholder value maximization as the main criteria for economic success, adopts a logical positivist stance (claiming to be 'value laden'), and searches for logical deductions based on unrealistic hypotheses (*instrumentalism*), and assumes that the whole is the sum of its parts (*reductionism*).

On the other hand, Ecological Finance Theory acknowledges the existence of complex feedbacks between the biophysical, production and

Table 1

Presuppositions of Neoclassical Finance vs. Ecological Finance Theory.

Presupposition	Neoclassical Finance Theory	Ecological Finance Theory
Financial markets	Efficiency <i>"I take the market efficiency hypothesis to be the simple statement that security prices fully reflect all available information"</i> (Fama, 1991, p.1575)	Decoupling <i>"Decoupling may arise from the growing influence of finance and the emergence of novel financial instruments"</i> Nyström et.al (Nyström et al., 2019, p. 575)
Money/ Banking	Loanable fund theory <i>"Neither banks nor do other intermediaries create loanable funds. That is the prerogative of spending units with surplus on income and product account"</i> (Gurley and Shaw, 1955, p.521)	Endogenous money <i>"Saving does not by itself increase the deposits or 'funds available' for banks to lend. Indeed, viewing banks simply as intermediaries ignores the fact that, in reality in the modern economy, commercial banks are the creators of deposit money"</i> Bank of England (2014, p.14)
Governance	Shareholder value <i>"How do we want the firms in our economy to measure their own performance? (...) the criterion should be maximization of the long-term market value of the firm"</i> (Meckling, 2001, p. 8)	Experimentation <i>"We need to overcome the tendency to recommend panaceas and encourage, instead, considerable experimentation at multiple levels to reduce the threats of massive collapses of valuable resources"</i> (Oström, 2012, p.128)
Modelling	Instrumentalism <i>"The proper test of a theory is not the realism of its assumptions but the acceptability of its implications"</i> Sharpe (1964, p. 434)	Realistic-ness <i>"When abstraction loses touch with reality, science becomes dogmatism"</i> Georgescu-Roegen (1971, p.319)
Ethics	Logical positivism <i>"It is well known that, under suitable assumptions (...) the allocation of resources in a competitive equilibrium is optimal in the sense of Pareto"</i> (Arrow and Debreu, 1954, p.265)	Moral science <i>"A society or an economy can be Pareto-optimal and still be perfectly disgusting"</i> (Sen, 1970)
Micro/macro link	Reductionism <i>"The firm is (...) a legal fiction which serves as a focus for a complex process in which conflicting objectives of individuals are brought into equilibrium within a framework of contractual relations"</i> . Jensen and Meckling (1976 p.311)	Holism <i>"(...) macroscopic properties emerge from local actions that spread to higher scales due to agents' collective behavior; these properties then feedback, influencing individuals' options and behaviors"</i> Levin et.al, (2012, p.113)

¹² Money, accounting and finance appeared after the Neolithic agricultural revolution, which allowed for the extraction of a regular caloric surplus (in excess of immediate needs of life reproduction) from the biosphere.

financial system. It seeks to evaluate the fairness and efficacy of a financial system based upon realistic assumptions (including those related to money creation). It adopts a *holist* perspective (i.e. acknowledging that ‘the whole is more than the sum of its parts’) and acknowledges that humankind is subject to ‘macrofoundations’ in the form of environmental, historical and societal forces that define and constrain modes of economic activity. It posits that claims of ‘value neutrality’ in finance are often the hypocritical articulation of a conservative bias, given that financial models, tools and instruments do not merely reflect reality but contribute to *shaping* it.¹³ It calls for empirically grounded theories, relying on relevant behavioral, social, institutional, and ecological assumptions (*realistic-ness*).

4. Metaphors for ecological finance theory

A truly ecological study of financial systems requires a completely new set of concepts. These concepts will foster the development of a new scientific agenda in finance, by allowing researchers to operationalize research questions, to develop and to test new research hypotheses. In order to reflect the interlocking of financial, productive and the Earth System (Fig. 2–4), these concepts draw on metaphors borrowed from biology. Table 2 lists these metaphors and compare them to their neo-classical finance counterparts.

Neoclassical finance takes *performance* (always narrowly defined from the investor or corporate perspective) as its core dependent variable. Ecological Finance Theory seeks instead to promote *resilience* by aligning financial models, tools, and policies with bio-geophysical and social constraints. Neoclassical finance considers *diversification* as a way to maintain value narrowly defined at the company level through financial metrics (such as ROA, EVA, and others...). Ecological finance theory evaluates the financial system from the perspective of *diversity* (which is a rather different concept, as explained farther below). Neoclassical finance theory posits that corporate *growth* is unambiguously beneficial. Ecological finance theory instead analyzes the implications of the limits of growth through concepts of *self-thinning* and *mitotic control*. Finally, neoclassical finance theory assumes that *financial transparency* is sufficient to generate an equilibrating and optimal vector of relative prices. Ecological finance theory calls instead for *ecological transparency* to incorporate the effects of economic activities on the biosphere and societies.

The next subsections introduce these new metaphors. We also discuss how they connect to each other, differ from metaphors in neoclassical finance, and reflect the new meta-hypotheses. These metaphors shall provide the basis of future theoretical and empirical examinations and explorations in Ecological Finance Theory.¹⁴ We also hope that they will permit a wider understanding among people from different disciplines.

4.1. Resilience

Resilience as an important concept in Ecological Finance Theory. It has been formally defined as the capacity of a system of any kind to

Table 2
Metaphors in Neoclassical Finance Theory and Ecological Finance Theory.

Neoclassical finance theory	Ecological Finance Theory
Performance	Resilience
Diversification	Diversity
Growth	Self-thinning and mitotic control
Financial transparency	Ecological transparency

¹³ In the words of Donald MacKenzie (borrowed from Milton Friedman), financial theory is « an engine, not a camera » (MacKenzie, 2006)

¹⁴ Note that other similar metaphors of biology applied to the economy realm can be found in Martínez (2018).

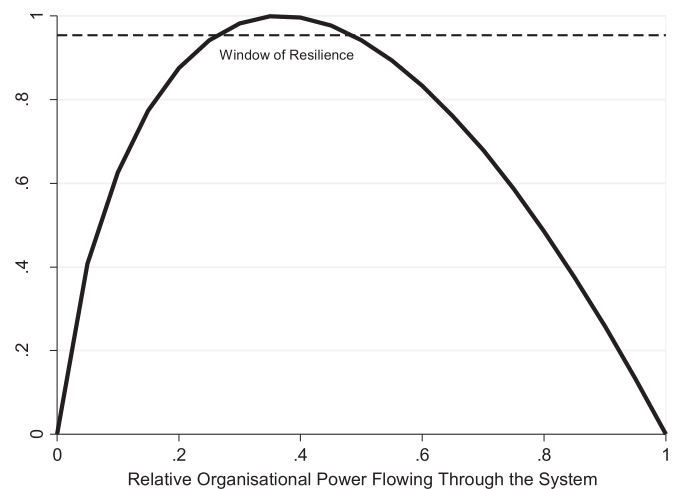


Fig. 6. Fitness for evolution and the window of resilience.

Note: this chart plots system fitness for evolution $F = -ka \log(a)$ for different values of $0 < a < 1$. The dotted line indicates values superior to 0.954 and the window of resilience. See Ulanowicz et al. (2009), Ulanowicz (2009), Ulanowicz et al. (2009) for details.

“adapt or transform itself in response to unfamiliar, unexpected events and extreme shocks” (Folke, Biggs, Norström, Reyers, & Rockström, 2016). As shown in Ulanowicz, Goerner, Lietaer, and Gomez (2009) any system’s (e.g. biomass, electrons, information, goods or money) total capacity for development (C) is the sum of two components: ‘throughput efficiency’ (A) and ‘resilience’ (ϕ):

$$C = A + \Phi \quad (1)$$

In Eq. (1) A refers to the system’s ability to exercise sufficient directed power to maintain its integrity over time, and ϕ refers to its ability to withstand shocks and adapt itself when necessary. ϕ represents the reserve that allows the system to persist and is the key to system’s resilience.

Defining $a = \frac{A}{C}$ as the relative measure of organizing power flowing within the system, and noting that a is bounded between 0 and 1, Ulanowicz et al. (2009) defined a system’s ‘fitness for evolution’ F using Boltzmann’s formulation:

$$F = -k \log(a) \quad (2)$$

In eq. (2) k is a positive scalar. If $a=1$ (all Capacity depends on Efficiency), F tends towards zero because the system has insufficient resilience. If $a=0$, then F also tends towards zero as the system has insufficient throughput efficiency. Finally, F goes up and down for intermediate values of a and peaks at $a = \frac{1}{e} = 0.36$. When $a < 0.36$, the system likely requires more efficiency. Conversely, when $a > 0.36$, the system may require more resilience. As shown in Fig. 6, there is a zone of optimal robustness, where $F > 0.95$. This zone, which has been observed in real ecosystems, has been named the “window of viability”. The figure is skewed to the right, indicating that in nature, fitness for evolution requires more resilience than efficiency. The neoclassical finance paradigm ignores resilience and focuses exclusively on growth of returns, profits, and other monetary signals (at the financial, macroeconomic, sectoral, corporate level). Ecological Finance Theory adopts instead a holistic perspective and acknowledges that resilience is an evolutionary constraint. Future research shall operationalize these concepts at various scales for framing new research questions, hypotheses and tests in Ecological Finance Theory.

4.1.1. Diversity

In nature, the resilience of each ecosystem is proportional to the diversity of species and the diversity of interactions among them. The

metaphor of *diversity* has several ramifications, which could provide the ground for a vast ecological finance research agenda. As shown in Table 3, the diversity of organizations and the diversity of their interaction in the socioeconomic sphere mirrors the diversity of biological species and relationships observed in an ecosystem.

Each natural ecosystem is characterized by a *biocenose*, i.e. a *complex system* made up of a web of species governed by individual behavioral norms, and interconnected through diverse forms of relationships. These relationships take various forms, including *predation* (e.g. a ladybird feeds from 300 to 1500 aphids during its lifetime) *parasitism* (e.g. lice on a human scalp), *commensalism* (e.g. the remora fixes its dorsal sucker to a more efficient swimming fish, which carries it to areas abundant in food); *synergies* (e.g. spruce trees grow faster with an undergrowth of broom given that the latter fixates azote), *mutual help* (e.g. herbivores of different species in Africa graze side by side in order to signal to each other the presence of carnivores), or *symbiosis* (e.g. the microbe flora obligated to live in human intestines help it to digest and protects it against pathogens or dysfunction). The relationship between the human species potentially includes all forms of relationship; however, *parasitism* has arguably become the dominant one (given the man-induced loss of biodiversity).

The diversity observed in nature mirrors institutional diversity in the economy. As shown in Table 3, five types of relationships characterize circulation within or between communities: gift, reciprocity, non-market exchange, and market exchange, which includes non-capitalist and capitalist market exchanges (Lagoarde-Segot & Paraque, 2018). Diversity is also evidenced by the existence of many different organizations with varied statuses and sizes (charity, limited liability company – LLC, public limited company – PLC, cooperative, mixed association, small and medium-sized enterprises (SMEs), etc.). Interactions between these entities take very different forms depending on place and time (Descola, 2005; Graeber, 2011; Meiksins-Wood, 2009, 2013, 2014; Ostrom, 2005; Servet, 2012; Testart, 2007).

Diversity appears a relevant and multifaceted metaphor with which to study the conditions for socioeconomic *resilience*. This metaphor allows us to study corporate governance and value creation from a macro-systemic perspective. It is fully aligned with the presupposition of

experimentation as a basis for firm governance (as opposed to shareholder value) (Table 1).

Diversity can be usefully contrasted with *diversification*; defined as a method to stabilize monetary portfolio value based on the mean, variance and correlation of cash flows of assets held by an investor. Diversification, indeed, stems from a representative agent's expected optimization behavior (usually a quadratic utility function) and excludes feedbacks to and from the Earth System from the analysis. By focusing on diversity, ecological finance research acknowledges that financial metrics are no more than the “*subjective construction of human beings who create and sustain a world of inter-subjectively shared meaning*” (Ardalan, 2008, p.7). Diversity is at odds with reductionism and instrumentalism, which reduces the company and its constituents to a set of atomistic agents adopting profit-maximizing behavior and hinders our understanding of top-down causation of macroscopic properties. The *diversity* metaphor is thus an invitation to revisit the notion of ‘value creation’ based on principles of realistic-ness, holism, experimentation, and in line with the overall objective of maintaining *resilience* (also connected with availability of resources).

4.1.2. Self-thinning and inequalities

In nature, *resilience* and *diversity* are both incompatible with the unlimited growth of individual species within an ecosystem. Growth indeed imposes constraints at individual, population, community and ecosystem levels. Biological research has shown that a trade-off exists between size and density of individuals sharing a common space. This trade-off has been empirically described as a ‘*self-thinning law*’ (or ‘*3/2 Power law*’) which posits that in denser populations of individuals, each one will have smaller average size, because space (or other resources) are limiting. For example, a forest of trees will have smaller trees if the planting density is very high (Fig. 7). However, the *self-thinning law* is broken in ecosystems with large inequalities in individual size. These inequalities allows the possibility of obtaining or computing larger averages of individual size than those expected for a given density. This effect was recorded for an intertidal marine algae (Fig. 8). In this algae very high density can coexist, at the same time, with large averages of size. But this was possible only because their size inequalities are extreme (recorded inequality Gini coefficients were between 0.9 and 0.95). That is, a single individual of this algae, in a given space, can effectively account for almost 95% of the entire mass of individuals in the whole population (Martínez & Santelices, 1992). In this biological species, observed individual growth could go beyond its limits only if inequality in the population was extremely high, with almost all plant biomass concentration in one single individual. It means that the computed average size does not exist in real individuals. It is only a mathematical result. What it really exists is a huge size inequality among individuals.

The *self-thinning law* is illustrated with Fig. 7, which plots log biomass (log B) per unit area vs. log density (log D) of survivors, known as the B–D diagram together with the outlier. This outlier has a higher mean size (biomass) than the expected values (those below the limiting slope). This case is only observed when inequality in individual size is extremely high. But in such populations, such high “mean-sized individuals” do not really exist: the computed average is only a mathematical illusion resulting from averaging sizes of highly unequal individuals.

The self-thinning law provides a useful analogy for understanding the conditions for resilience and diversity in the socioeconomic sphere. The Global Production Ecosystem (GPE) is made of a very dense network of transnational corporations (TNCs) that link production ecosystems globally through their subsidiaries (Österblom et al., 2015). The development of TNCs in turn goes hand in hand with an unprecedented concentration in global ownership. In a major study covering over forty thousand transnational corporations, Vitali et al. (2011, 2014) demonstrated that transnational corporation ownership and control is extremely concentrated. A growing network of interconnected financial

Table 3
Diversity.

In the biosphere	
Predation	One species uses another specie (prey) for food consumption
Parasitism	One species uses another specie to feed and to reproduce without inducing its death
Commensalism	One species survive thanks to the presence of another species to which it is indifferent
Synergy	The development of an organism is stimulated by the presence of another species
Mutual help	Living organisms associate with one another in order to reach a shared benefit (even if they could develop independently in normal conditions)
Symbiosis	Two protagonists benefitate from reciprocal advantages and could not survive without the union.
In the socioeconomic sphere	
Gift	Gift exchange strives to be disinterested and timeless. There is no inherent obligation or right, either for the transferer or the recipient. It remains unilateral.
Reciprocity	It implies neither synchronicity nor obligation in its inherent two-way movement.
Exchanges	They concern a causal relationship and an obligation.
Non-market exchange	Price or value considerations are not sufficient to provoke the exchange.
Non-capitalist market exchange	The exchanger seeks use value from the good provided in return.
Capitalist market exchange	The exchanger seeks exchange value from the good provided in return.

Source: Testart (2007); Lagoarde-Segot and Paraque (2018).

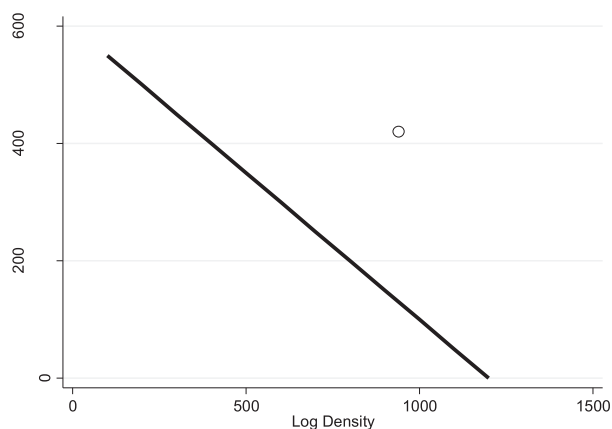


Fig. 7. Self-thinning and the B–D diagram.

Note: this figure displays the Logarithmic relationship between a measurement of mean size of individuals of a population as it decreases at higher density in a given space. For plants the slope is -1.5 and it has empirical limit extreme values at both axes, related to architectural constraints given by forms. The point over the limit slope represents an unexpected combination of size and density, only found mathematically possible when there was a huge size inequality among the observed individuals (Martínez & Santelices, 1992).

intermediaries controls the network: financial intermediaries hold shares larger than 5% in 13% of non-financial companies and in 60% of other financial companies. A few financial actors dominate all segments of production, control the whole supply chain and have a disproportionate influence on decision-making (Clapp & Fuchs, 2009). These actors are marked by a high level of size, internationalization, complexity, interconnectedness, and lack of substitutability (in providing key services, such as market infrastructures). Their distress or disorderly failure would cause significant disruption to the wider financial system and economic activity. Fig. 8 develops this analogy by comparing the global network of corporate control with the marine alga of Fig. 4, where one individual almost represents ‘the whole forest’. Ecological Finance Theory seeks to carve out strategies that could enhance *self-thinning*, in order to foster *diversity* and *resilience* in the socioeconomic sphere.

4.1.3. Mitosis, self-regulation and growth

Neoclassical finance theory urges endless growth because it implicitly assumes that growth is a sufficient measure of health. Growth or growing is indeed an innate goal that defines every living organism. However, one fundamental principle of biology is that each individual and population have growth limits (Fig. 9). A bacterial cell, and all cell types, divides in two through a mechanism called *mitosis*. Then, each daughter cell soon grows to the same size as the cell that gave rise to it, then remains unchanged. What grows is the number of cells, as the growth of different organs in an individual. But even this is limited. For example, the cells that make up an organ (liver, skin, muscles, the bones) all possess information allowing them to reach a certain size and then stop. The whole body also reaches a limited size (at dynamic equilibrium). Excesses of size are always signs of illness, obesity or cancer to name a few. Unlimited growth finds detention at the depletion of resources (lack of food for the case of bacterial growth) or exhaustion of the body’s energy in a patient with cancer, and the resulting body degradation and death. It is easy to imagine how this metaphor of unlimited growth exhausting the system also applies to the connections between the biosphere and the economic world.

Limitless expansion decouples production from the natural processes required to sustain the desired production outcome. For instance, the use of human input (e.g. fertilizers) has suppressed short-term variance in harvestable biomass and in consumer prices, resulting in higher growth. But it also suppressed important feedback signals and decreased the resilience of global and local ecosystems (Nyström et al., 2019). In the case of chemical fertilizers, harvests increase in the short term but in the long term soils die by excess of salts. A global homogenization of species (including genes, varieties and crops) the replacement of tropical forests by spatially extensive monocultures (for instance, soy and oil palm plantations), and the homogenization of rainforest bacterial communities, as a result of land conversion to cattle pasture, have also been observed. All these factors are documented in the scientific literature and are detrimental to diversity and to *resilience* (Jefferson, Köllhofer, Ehrlich, & Jefferson, 2015; Koh and Wilcove, 2008; Rodrigues et al., 2013).

As a further metaphorical analogy, Fig. 9 juxtaposes a bacterial mitosis process with the Great Acceleration of GDP since 1750. Collapse can be easily suspected to come soon, on both cases because resources do have limits (for humans, social collapse can be foreseen too). Biological systems have included the internal control to growth, probably as an evolutionary retained strategy to keep alive. As the time of writing, the

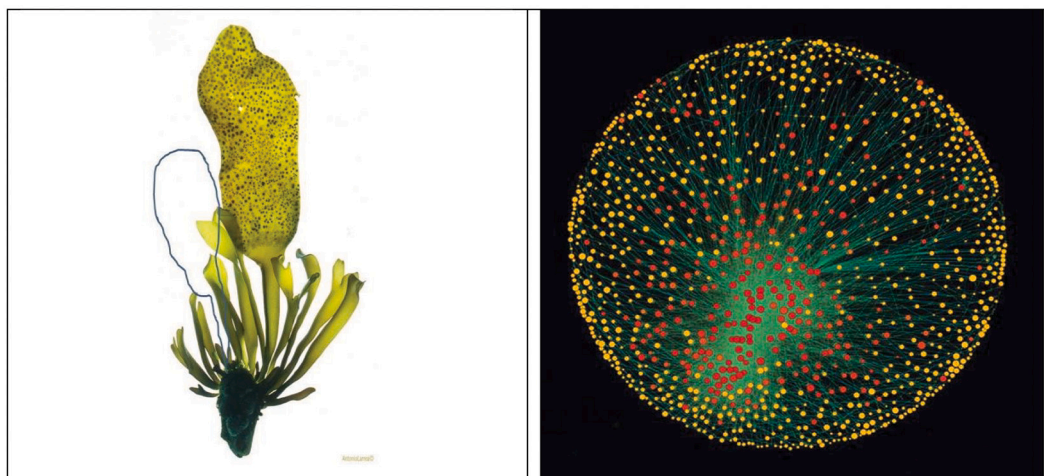


Fig. 8. Self-thinning and inequalities.

LEFT: The contours of a “mean sized alga” to the left of the big individual in the center is the schematic representation of an often non-existent but statistical “mean individual”. As a metaphor of the non-existing mean income (GDP) in the economy of many countries where inequalities are dominant. RIGHT: Dark dots indicates a super-connected firm, a light dot indicates a very connected firm. The size of each dot is proportionate to the firm’s revenues. The high level of observed interconnectedness suggests that the ‘mean sized’ multinational corporation does not exist. Source: Vitali et al. [2011].

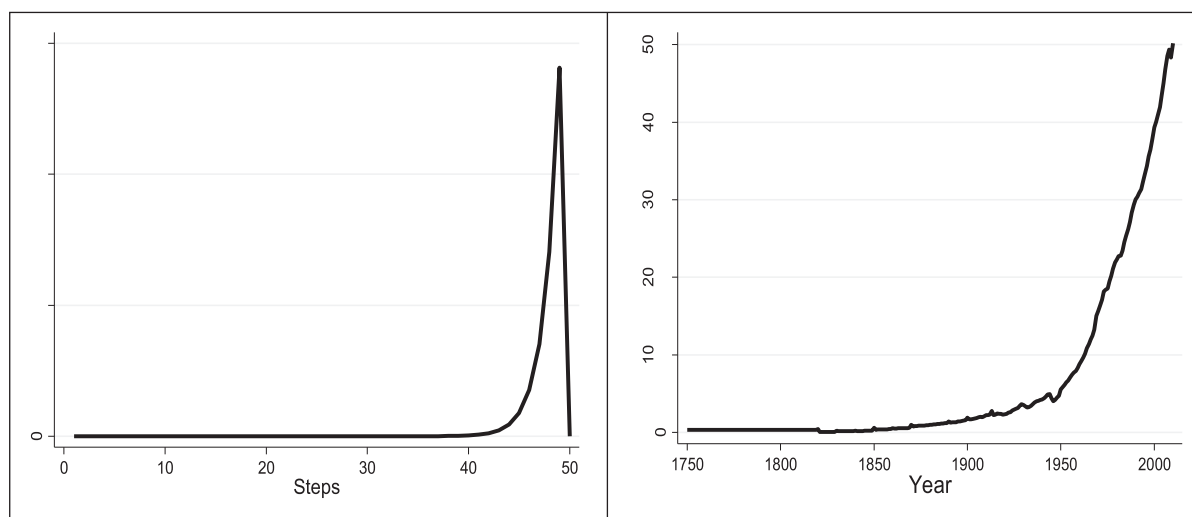


Fig. 9. Mitosis process and global GDP per capita.

Note: this figure compares the exponential evolution of world GDP per capita and that of a cell doubling in numbers following less than fifty steps of mitosis (doubling at each step or time interval common in cellular type of growth). The data on the right-hand side of the figure is taken from the Maddison project (2018).

COVID-19 pandemic – a biological shock – has been touted as the largest recession in modern economic history. It remains to be seen whether the GDP per capita curve will eventually collapse and mirror the *mitosis* curve on the left-hand side of Fig. 9, and validate the doom predictions of the 1972 Meadows Report of the Club of Rome.¹⁵

4.1.4. Ecological transparency

A system is better understood (and its limits better recognized) if the functioning of all components is transparent to all those observing (and using) the system. *Transparency* is of course an important metaphor in neoclassical finance. However, such transparency is bounded by the information provided by financial markets and accounting systems. Financial transparency seeks to preserve the value of financial capital, but excludes the two other types of capitals necessary for production (human capital and natural capital). *Ecological transparency* extends financial transparency to the effects of business on society and the Earth System. Ecological Transparency requires inventing new methods for calculating costs, taking into account not only the preservation of financial capital, but also the preservation of natural and social capital.

The triple depreciation line approach (TDL) (currently tested in France as part of the CARE¹⁶ project) is one such method. Under this new system, a company's balance sheet includes the used-up natural and human capital as charges and records corresponding user costs as assets (Table 4). Amortization costs associated to the use of natural and human capital resources then appear in the income statement. This leads to a new definition of profits reflecting the real costs of activities (Table 5). The triple depreciation of tangible, natural, human capital may thus preserve limited resources, in line with strong sustainability (Table 6).

Under this system, polluting firms would purchase natural capital goods supplied by the ecological service providers in order to amortize the depreciation of natural capital induced by their activities. For instance, a paper plant would hire the services of another firm as part of its amortization charges in order to clean polluted freshwater.

Ecological transparency is a condition for *resilience*. If something is ignored (or hidden), the understanding will be partial and the managing will be more difficult. For instance, in human biology if one hides from a medical doctor the symptoms or day-to-day behaviors, it will be more

Table 4

Balance sheet (CARE/ triple depreciation line).

Assets		Liabilities	
Cash	100	Equity	100
Human capital user costs	100	Human capital	100
Natural capital (user costs)	50	Natural capital	50
Total	250	Total	250

Source: adapted from Rambaud and Richard (2015).

difficult for the doctor to control the illness. Ecological transparency should also help controlling client fidelity, it contributes to better and sustainable relationships within and between organizations. It could also help consumers to take intelligent and healthy decisions for them and for the whole system by taking ecological aspects into full consideration (Martínez & Olguín, 2013).

5. Puzzle solving: biology-sourced concepts and economic modelling

The scientific objective of Ecological Finance Theory is to *align financial models, tools, and policies with social and biophysical constraints*. This implies articulating our new hypotheses and concepts in a robust analytical framework. To do so, we introduce *Philia 0.1*,¹⁷ the simplest theoretical stock-flow consistent (SFC) model we could think of. *Philia 0.1* is comprised of 45 accounting and behavioral equations interlocked in a sectoral transaction matrix. It permits to build informed intuition on the effects of *ecological transparency* on *diversity* and *resilience*. It also employs realistic mechanisms of *endogenous money* in an analytical framework allowing for *mitotic control*. In what follows, we first describe the model, and then analyze convergence to a stationary steady state, following the introduction of an ecological transparency shock. In our setting, this shock occurs through the depreciation of natural capital by brown sector firms, the rediscounting of loans by an Ecological Central Bank, and increased consumer preference for ecological sector goods.

¹⁷ *Philia* is a Greek word meaning friendship, affective proximity, goodwill and positive expectations. Ecological Finance could be seen as a way to amend Marx's general formula of capital as follows: *Philia - M - C - P - C' - M' - Philia*, i.e. embedding capitalistic accumulation within the accumulation of a global ethical and ecological value (Arvidsson, 2009; Paraque and Revelli, 2017).

¹⁵ See <https://www.clubofrome.org/>

¹⁶ See <https://www.chaire-comptabilite-ecologique.fr/Natural-capital-visibility-in-financial-accounting-Method-3-Extended-Version>

Table 5
Income statement (CARE/ triple depreciation line).

Gross profits	300
Depreciation:	
Financial	−100
Human capital	−100
Natural capital	−50
Net common profit	50

Source: adapted from [Rambaud and Richard \(2015\)](#).

Table 6
Final balance sheet (CARE/ triple depreciation line).

Assets		Liabilities	
Cash	$100 - 100 + 100 = 100$	Financial capital	100
Human using costs	$100 - 100 + 100 = 100$	Human capital	100
Natural assets (using costs)	$50 - 50 + 50 = 50$	Natural capital	50
Common profits	50	Common net profits	50

Source: adapted from [Rambaud and Richard \(2015\)](#).

These events trigger a *qualitative* transformation of the economy marked by higher systemic *diversity* and *resilience*.

5.1. Meta-theoretical hypotheses

The stock-flow consistent (SFC) model literature developed recently following the groundbreaking work of [Godley and Lavoie \(2012\)](#). This modelling approach rest on principles fully aligned with several hypothesis and presuppositions of Ecological Finance Theory. *First*, SFC models break away with *reductionism* and propose a *systemic* representation of the economy. This representation based on a *realistic* accounting framework integrating the real and the financial sides of the economy, as well as their impact of natural systems. *Second*, SFC models depart from *instrumentalism* and offer an analytical framework in which “there is no need or room for the rational expectation hypothesis” ([Godley & Lavoie, 2012](#), p.16). *Third*, SFC modelling permits to account for *diversity*, as they do not rely on neoclassical foundations of utility maximization principles or the availability of perfect information. They instead retain a principle of adjustment to observed disequilibria, by positing that economics agents display *procedural rationality*, i.e. “set themselves norms and targets, and act in line with these and the expectations that they may hold about the future” (ibid, p.16). *Fourth*, SFC models acknowledge the endogeneity of money and permit to track the creation, circulation and destruction of inside and outside money. *Fifth*, a stationary steady state (SSS), in which flows and stocks variables remain in a constant relationship to each other, while also being stable in levels, can be simulated (or calculated) for reasonably-sized models. This notion mirrors the concept of *mitotic control*. *Sixth*, the SFC approach can be used in various settings, including theoretical analysis ([Mazier, 2020](#) and the present model), in conjunction with econometric modelling ([Gimet and Lagoarde-Segot, 2019](#)), or for prevision purposes ([Dafermos and Nikolaidi, 2018](#)). SFC models shall thus play a significant role in the future development of Ecological Finance Theory.

5.2. The model's transaction flows matrix

The model's transaction matrix is shown in [Table 7](#). This transaction matrix constitutes the accounting backbone of the model and ensures that all flows and stock readjustment are interlocked in a watertight accounting structure integrating the real and financial sides of the economy. The matrix describes a simplified economy with five institutional sectors: households, production firms (divided into the sustainable and the non-sustainable sector), a banking sector and a Central Bank. For simplicity, there is no specification of a particular kind of government in the model. This assumption (which can also be

interpreted as the government maintaining a strictly balance budget) implies that money is only emitted through bank loans (i.e. there is no ‘outside money’ in the model).¹⁸

In line with double entry bookkeeping principles, each economic transaction is double-sided, i.e., one sector's income is another sector's spending, and one sector's asset is another sector's liability. Changes to the stock of assets and liabilities (in columns) result from the budget constraint of each individual sector, which itself depends on annual flows of income and spending (in rows). This ensures that there are no black holes in the system (*‘everything comes from somewhere and goes somewhere’*) and that the real and financial sides of the economy are fully integrated. Following SFC modelling convention, a (+) sign indicates a source of funds while a (−) sign indicates a use of funds.

Column 1 shows the budget constraint of households. Households receive income as wages (W), profits redistributed by firms (F) and banks (FB), and interests on their deposits ($r_{d(-1)}L_{(-1)}$). Households spend their income on consumption expenditures (C). At the end of each period, excess of income over spending accumulates in stocks of banking deposits (ΔM).

Column 2 and 3 divide the production firm sector's activities into a current and capital account. These columns amalgamate the accounts of ‘ecological’ and ‘brown’ firms. Column 2 shows the receipts and outlays of the production sector. This sector receives payment flows on sales of final ecological goods (C_e) and brown goods (C_b). It also receives income on sales of ecological goods (I_e) and brown capital goods (I_b). Both ecological and brown firms pay wages to households ($W_e + W_b$), interest on their loans in euros ($r_{l, e(-1)}L_{e(-1)} + r_{l, b(-1)}L_{b(-1)}$) and set aside funds for depreciation allowances ($DA_e + DA_b$). Column 3 shows that firms' capital expenditures are financed by depreciation allowances (DA) and by new loans ($\Delta L_{d, e} + \Delta L_{d, b}$). In our simple setting, profits ($F_e + F_b$) are fully distributed to households.

Columns 4 and 5 show the current account and the capital account of banks. Column 4 shows that banks receive interest payments on loans ($r_{l, (-1)}L_{(-1)}$) and pay interests on deposits in euros ($r_{d(-1)}L_{(-1)}$). Column 5 shows adjustment to banks' simplified balance sheet, with new deposits (ΔM) on the liability side and private sector loans (ΔL_d) and high-powered money (ΔA) on the asset side. Net equity corresponds to the difference between assets and liabilities. It is entirely redistributed to the household sector as dividends (FB).

Column 6 shows the capital account of the Central Bank. In our simplified setting, the Central Bank's only liability is the high-powered money (reserve currency) that it issues (ΔA) by rediscounting a fraction of bank loans ($\Delta \phi$).

5.3. The model's equation

5.3.1. Macroeconomic variables

We begin with the description of total gross production (GDP). GDP, denoted Y is defined as the sum of all expenditures on goods and services, including consumption C and investment goods I in the ecological sector (e) and the brown sector (b):

$$Y = \sum_{i=e,b} (C_i + I_i) \quad (1)$$

In line with national accounting, GDP must also be equal to the sum of factor payment in the economy, which consists in the wage bill (WB), production firms profits (F), banking profits (FB), and the depreciation and amortization funds (DA) that ecological and brown sector firms set aside to replace used-up capital.

¹⁸ As explained by the Bank of England (2014), and in much of the Post-Keynesian economics literature ([Lavoie, 2015](#)) the majority of money circulating in the real economy is issued this way.

Table 7The accounting transactions flows matrix of model *Philia 0.1*.

Transactions	Households	Production firms		Banks		Central Bank
	(1)	Current (2)	Capital (3)	Current (4)	Capital (5)	Capital (6)
Consumption	$-C$	$(C_e + C_b)$				
Investment		$(I_e + I_b)$	$-(I_e + I_b)$			
[GDP]		Y				
Wages	$+W$	$-W$				
Depreciation allowances		$-\left(\begin{smallmatrix} DA_e \\ +DA_b \end{smallmatrix}\right)$	$-(DA_e + DA_b)$			
Interest on loans		$-(r_{l,e(-1)}L_{e,-1} + r_{l,b(-1)}L_{b,-1})$		$+r_{l(-1)}L_{-1}$		
Interest on deposits	$r_{d(-1)}L_{(-1)}$			$-r_{d(-1)}L_{(-1)}$		
Changes in loans			$\Delta L_{d,e} + \Delta L_{d,b}$		$-\Delta L_d$	
Changes in deposits	$-\Delta M_m$				ΔM	
Firm profits	$+F$		$-(F_e + F_b)$			
Bank profits	$+FB$				$-FB$	
Changes in High-Powered Money					$+\Delta A$	$-\Delta A$
Loan rediscounting					$-\Delta \varphi$	$+\Delta \varphi$

$$Y = WB + F + FB + \sum_{i=e,b} (DA_i) \quad (1.2)$$

Total personal disposable income, YD is defined as the difference between GDP net of depreciation and amortization funds:

$$YD = Y - \sum_{i=e,b} (DA_i) \quad (3)$$

As shown in (4) and (5), depreciation and amortization funds are a constant proportion $0 < \delta < 1$ of the stock of capital K_{-1} that firms hold at the beginning of the fiscal year. Both ecological and brown sector firms set aside funds to preserve the stock of tangible capital (δ_k). Following the ecological transparency shock, brown sector firms will set aside additional funds to account for the depreciation of the consumed natural capital assets (δ_N). In the initial stationary-steady state of the model however, $\delta_N = 0$.

$$DA_e = \delta_k K_{-1} \quad (4)$$

$$DA_b = (\delta_k + \delta_N) K_{-1} \quad (5)$$

The wage bill (WB) is equal to the product of the wage rate W times the level of employment N_d . N_d is determined based on output Y and labor productivity pr :

$$WB = W \cdot N_s \quad (6)$$

$$N_d = Y / pr \quad (7)$$

$$W = WB_d / N_d \quad (8)$$

5.3.2. Households and consumption

Total household consumption (C_d) is equal to the sale of final goods by ecological (C_e) and brown sector firms (C_b) (Eq. 9). Household demand for consumption goods is split in the two sector according to a parameter ($0 < \theta < 1$) reflecting consumer preferences among existing alternatives (Eqs. 11 and 12). This parameter applies to a dynamic consumption function, which includes an autonomous component (α_0), disposable income (according to a factor α_1) and the stock of banking deposits inherited from the previous period (according to a factor α_2) (Eq. 10). As shown in Eq. 12, brown sector firms amortize the depreciation of natural capital entirely by spending in the ecological sector.

$$C_d = \sum_{i=e,b} (C_i) \quad (9)$$

$$C_h = \alpha_0 + \alpha_1 (YD) + \alpha_2 (M_{h,m-1}) \quad (10)$$

$$C_b = \theta C_h \quad (11)$$

$$C_e = (1 - \theta) C_h + \delta_N K_{-1} \quad (12)$$

At the end of each year, whatever disposable income is not spent on consumption good C_d is added to the stock of bank deposits M_h :

$$\Delta M_h = M_h - M_{h-1} = YD - C_d \quad (13)$$

5.3.3. Firm investment

The demand for investment (I_d) is the sum of investment from ecological and brown sector firms. IN each sector, demand for capital goods based on the partial accelerator model. As shown in Eq. (14), the first component of gross demand for investment is forward looking and adjusts partially ($\gamma < 1$) to the discrepancy between the targeted capital stock K^T and the stock of tangible capital K_{-1} inherited from the previous period. The second component is equal to the investment required to replace the used up machines DA :

$$I_d = \sum_{i=e,b} (\gamma (K^T_i - K_{-1}) + DA_i) \quad (14)$$

Firms from each sector target the capital stock K^T based on macro-economic sales achieved in the previous period:

$$K^T_e = \kappa_e Y_{-1} \quad (15)$$

$$K^T_b = \kappa_b Y_{-1} \quad (16)$$

In each sector, the stock of capital is equal to the stock inherited from the past period net of depreciation allowances, plus newly acquired capital:

$$K = K_{-1} - \sum_{i=e,b} (DA_i) + I_d \quad (17)$$

The demand for new loans is equal to the net demand for investment, i.e. to the difference between gross demand for investment minus the amortization funds:

$$\Delta L_d = I_d - \sum_{i=e,b} (DA_i) \quad (18)$$

At each period, firm profits are equal to the difference between GDP and other factor payments:

$$F = Y - WB - FB - \sum_{i=e,b} (DA_i) \quad (19)$$

5.3.4. Banks

Banks behave passively, and open lines of credit to firms such that

the stock of loans increases with loan demand (eqs. 20 to 25). The stock of loans is equal to the stock of banking deposits held by households.

$$\sum_{i=e,b} L_{s,i} = \sum_{i=e,b} L_{d,i} \quad (20)$$

$$L_{s,e} = L_{d,e} \quad (21)$$

$$L_{s,b} = L_{d,b} \quad (22)$$

$$M_s = L_s \quad (23)$$

Banks profits are equal to the interest spread, plus the addition to their stock of reserve assets, held at the Central Bank's deposit facility (and which pay zero interest).

$$FB = r_{l,M}(L_{m,-1}) + -r_{d,M}(L_{m,-1}) + \Delta A_s \quad (24)$$

5.3.5. The 'Ecological' Central Bank

The Central Bank provides reserve on demand, up to the value of the collateral pledged by the bank requesting reserves. As in reality, money is therefore endogenous given that reserves therefore do not constrain lending in the economy (Lavoie, 2015; Bank of England, 2014; Lagoarde-Segot, 2020). However, the Ecological Central Bank issues reserves to the banking system up to the Social Return on Investment (SROI)¹⁹ of previous loans issued to the ecological sector.

Banks ask extra-financial rating agencies to compute the SROI of the stock of ecological sector loans issued at the previous period ($L_s, e-1$). This monetized value of impact is represented with the letter:

$$\varphi = (1 + SROI)L_{s,e-1} \quad (25)$$

Banks then turn to the Central Bank to convert of a fraction of these impact certificates into high-powered money. The Central Bank thus rediscounts a fraction τ of the ecological sector loan portfolio, in exchange for the SROI certificates. Letting τ denote the applied haircut rate; the amount of high-powered money issued (ΔA) is given by:

$$\Delta A_d = \tau \cdot \frac{\varphi_{t-1}}{(1 + SROI_{t-1})} \quad (26)$$

$$\Delta A_s = \Delta A_d \quad (27)$$

$$\Delta H_s = \Delta A_s \quad (28)$$

The Ecological Central Bank hence rediscounts ecological sector loans based on their expected positive impact on nature and society systems. This process is akin to a slow *metamorphosis* of the monetary system, in which the Central Bank issues high-powered money in exchange for ecological impact. This mechanism certainly departs from existing current monetary policy practices, but stands aligned with ecological finance presuppositions (such as *experimentation* and finance scholarship as a *moral science* (Table 1)). Money is indeed a temporary institution, defined as “the means by which societies gives back to each of its members what it judges each of them to have given to it” (Aglietta, 2018, p.32), or as a “relation of belonging (...) that links each member of the social group to the whole” (ibid p.6). We set $\tau = 0$ in the first calculation of the model's SSS, and then introduce the Ecological Central Bank with our ecological transparency shock.

¹⁹ SROI measures “how change is being created by measuring social, environmental and economic outcomes and uses monetary values to represent them. This enables a ratio of benefits to costs to be calculated. For example, a ratio of 3:1 indicates that an investment of £1 delivers £3 of social value. SROI is about value, rather than money. Money is simply a common unit and as such is a useful and widely accepted way of conveying value”. (Nicholls, Lawlor, Neitzer, & Goodspeed, 2009). The SROI methodology provides a powerful frame to construct worthiness and value creation for activities, even in cases where no actual financial returns are generated (Yates & Marra, 2017; Cooney, 2017).

5.3.6. Resilience

In line with section 4.1, we define total capacity for development (D) as the sum of two components: ‘throughput efficiency’ (Y) and ‘resilience’ (R):

$$D = Y + R \quad (29)$$

Resilience is given by a Michie and Oughton (2013) diversity index, which we apply to the components of macroeconomic output and sales:

$$R = 1 - \sum_{i=e,b} \left[\left(\frac{C_i}{C} \right)^2 + \left(\frac{I_i}{I} \right)^2 \right] \quad (30)$$

We compute $a = \frac{Y}{C}$ as the relative measure of organizing power. The system's ‘fitness for evolution’ F is finally given by eq. (31) (where $k = 2.71$):

$$F = -k \log(a) \quad (31)$$

5.3.7. Closure and parameters

For the model to be *stock-flow consistent*, the n^{th} equation must be redundant, that is to say, determined by the other $n-1$ equation. Our model's hidden equation is the savings-investment identity. The inner workings of the model imply that at each and every state of the model, the quantity of deposits held by households in circulation is strictly equal to the stock of loans implied by firms' demands and banks' balance sheets. This closure condition is given by eq. (32):

$$\Delta M_s = \Delta L_s \quad (32)$$

Finally, the model includes exogenous parameters described in eqs. (33) to (44)

$$r_{l,m} = \overline{r_{l,m}} \quad (33)$$

$$r_{l,c} = \overline{r_{l,c}} \quad (34)$$

$$SROI = \overline{SROI} \quad (35)$$

$$\delta_{e,b} = \overline{\delta_{e,b}} \quad (36)$$

$$\alpha_0 = \overline{\alpha_0} \quad (37)$$

$$\alpha_1 = \overline{\alpha_1} \quad (38)$$

$$\alpha_2 = \overline{\alpha_2} \quad (39)$$

$$\theta = \overline{\theta} \quad (40)$$

$$\kappa_{e,b} = \overline{\kappa_{e,b}} \quad (41)$$

$$\tau = \overline{\tau} \quad (42)$$

$$\delta_K = \overline{\delta_K} \quad (43)$$

$$\delta_N = \overline{\delta_N} \quad (44)$$

5.3.8. Mitotic control: the stationary steady state

We first calibrate the model's parameters without ecological transparency (setting $\delta_N = \tau = 0$). We then bring the economy to a *stationary steady-state* (SSS) defined as a state where stock and flow variables remain in a constant relationship to each other, while also being stable in levels. The SSS is rarely reached in the real world where functional relationships, parameters and exogenous variables are changing all the time. As shown in Godley and Lavoie (2012), the SSS nonetheless serves as a useful analytical device permitting to gain insight into the impact of the initial and terminal effects of a given shock to the economic system. Referring to the SSS is particularly relevant in the context of ecological finance theory, as it ensures the modelled dynamic respects conditions

of *mitotic control*. The model reaches a stationary state in which there are no new stocks and flows, and where the value of GDP is 2.739 after 40 replications.

Using this stationary state as our baseline scenario, we then introduce three *ecological transparency* shock. *First*, brown sector firms set aside one third of their capital stock as natural capital amortization charges, following the implementation of a TDL accounting system. We therefore increase the value of δ_N from 0 to 30% (as shown in eq. (12), brown firms will spend these depreciation funds entirely in the ecological sector). *Second*, an Ecological Central Bank supports the ecological transition by rediscounting up to 50% of the SROI value of bank loans to the ecological sector ($\tau = 50\%$). *Third*, we assume that increased transparency triggers an increase in household demand for ecological sector goods. We therefore decrease the value of θ by 10% in (from 0.9 to 0.81) (Eq. 11).

Following the introduction of these shocks, the models reaches a SSS after 20 replications, with GDP taking a terminal value of 4.02. Figs. 10 and 11 report the evolution of key variables during the transition towards the new SSS.

5.3.9. Analytical results

Fig. 10 displays the response of the macroeconomic system to the shock. Inspection of the figure indicates that accounting consistency is respected throughout the simulations. The upper left figures show the GDP and its income components (Eq. 1.2). The upper-right figures shows GDP and its spending components (Eq. 1). The lower-left figure validates the model's consistency. It shows that the difference between Eq. (1) and Eq. (1.2) is zero ('accounting' line). It also shows that Eq. (32) is respected throughout the simulations, as the quantity of deposits held by households in circulation is equal to the stock of loans implied by firms' demands and banks' balance sheet ('closure' line). The figure in the lower-right quadrant displays the evolution of the annual growth rate of macroeconomic consumption. This figure reveals a modest increase in

total consumption. However, the latter stems from a sizeable increase in the consumption of ecological goods, associated to a drop in the consumption of brown goods. A *qualitative* transformation in the economy is therefore taking place. This transformation also appears compatible with a controlled growth (*mitotic control*) of aggregate consumption.

Fig. 11 provides further details about the economy's reaction to the shock. The upper left figure shows that ecological transparency increases the system's overall *fitness for evolution* (Eq. 31). This is due to a simultaneous increase in *resilience* (Eq. 30) and a drop in *organizational power*. The upper right figure indicates that the shown increases brown sector firms' amortization charges. These additional expenses correspond to new sales for the ecological sector, as can be seen from a sharp increase in ecological consumption. The same figure also shows that depreciation charges remain constant throughout the simulation for ecological firms. A *qualitative* transformation of the economy is again taking place, given that internalization of natural and human capital externalities by brown firms funds the expansion of the ecological sector. One could think of Google, Amazon or Microsoft amortizing the depreciation of natural capital induced by their activity by funding resilient organic food systems.

The lower left figure shows how the banking sector reacts to the shock. Banking profits remain constant; however, the introduction of the Central Bank's rediscounting scheme (Eqs. 25 to 28) leads to a significant increase of the reserve ratio. This policy is therefore compatible with *mitotic control* of banking sector development, and appears conducive to financial stability. Finally, the lower right figure shows the evolution of macroeconomic investment. While the ecological sector's investment increases, the brown sector is behind much of the observed increase in investment. However, this figure measures gross investment, i.e. taking into account natural capital amortization charges (Eq. 14). The observe dynamic hence suggests that during the transition to a new SSS, many brown sector firms may join the ecological sector, after investing sufficient resources in their own ecological technologies and

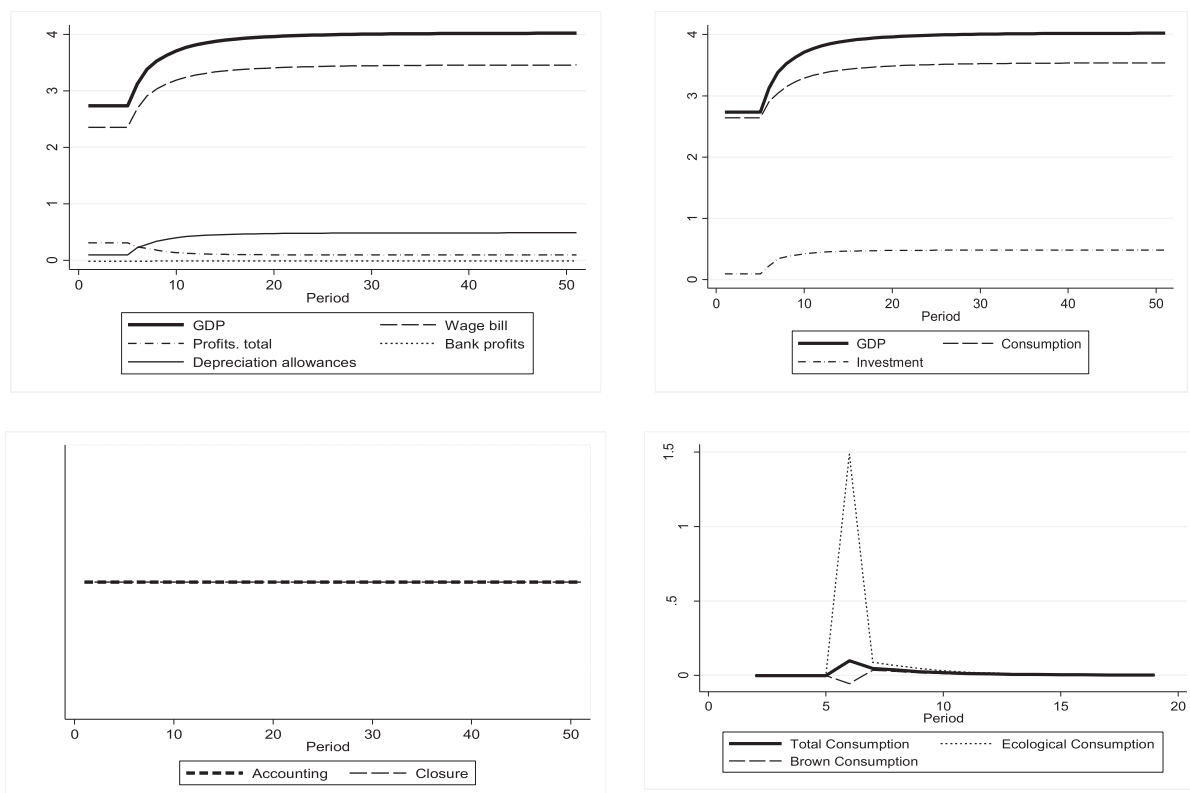


Fig. 10. Model simulation.
Source: Authors' simulation.

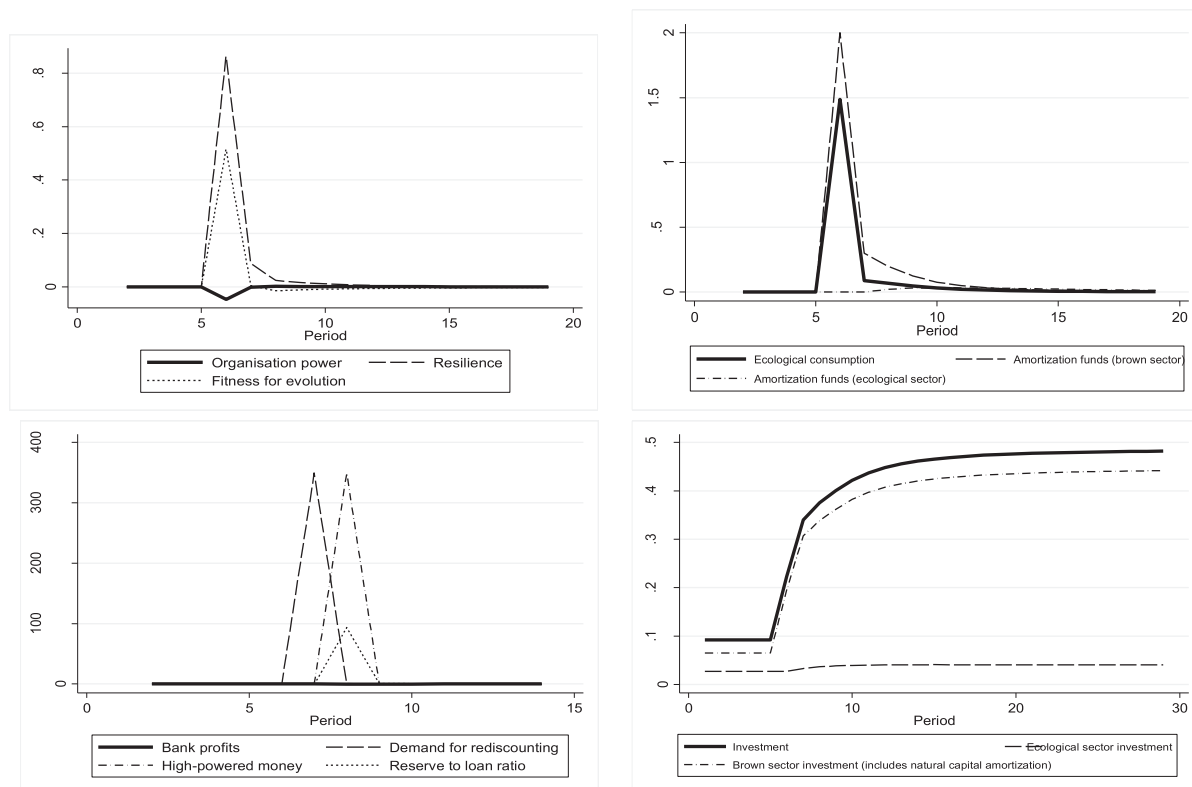


Fig. 11. Model simulation.
Source: Authors' simulation.

production processes. However, the present model does not account for such a mechanism.

6. Conclusion

A growing consensus among various disciplines and stakeholders indicates that biological systems, socio-economic systems, and financial systems can no longer be studied separately in the 21st century. Unfortunately, modern finance theory, by virtue of its meta-theoretical hypotheses *superimposes financial logics to the analysis of the Earth System*, when addressing the new context would instead require *the order to be turned upside down*. The ecological, social and economic challenges of our generation necessitate the revisiting of financial models, systems, structure, policies, instruments, and governance.

This paper has proposed *Ecological Finance Theory* as a new vision for 21st century financial economics. We have attempted to delineate this theory by reference to the three epistemological layers of discourse that structure scientific activities. We have introduced a set of comprehensive *meta-theoretical hypotheses*, new research *metaphors*, and *puzzle-solving tools*, which we have compared and contrasted with those used in the neoclassical finance paradigm. In particular, we have introduced biological concepts of *resilience*, *diversity*, *self-thinning*, *self-regulated mitosis*, and *transparency* in the field of financial economics. We then analyzed the impact of an increase in ecological transparency on *resilience*, *diversity* in the context of a new stock-flow consistent model with *mitotic control*.

We believe that this paper contains the basic elements of a new conceptual and methodological apparatus in our discipline. This new apparatus should allow our community to incorporate the complex feedback systems between financial systems, social provisioning and Earth systems into our analyses and prescriptions.

While we have sought to describe and contribute to a budding scientific revolution in the field of finance, Ecological Finance Theory

nonetheless remains work in progress. It can only succeed in supplanting the neoclassical paradigm if it attracts sufficient attention from finance scholars (especially junior researchers and students) and practitioners. The immediate task ahead for interested scholars may lie in operationalizing the content of this paper (for instance, by introducing new datasets, research questions, theoretical investigations and empirical modelling). We hope that new ecological finance metaphors, research questions, tools, puzzles, paradoxes and controversies will quickly emerge. We invite readers to join the discussion, informally by email, or in varied outlets, such as the Post-Crisis Finance PoCfN Working Papers Series and the *International Review of Financial Analysis*.

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References

- Aalbers, M.B. 2015. The International Encyclopedia of Geography: People, the Earth, Environment, and Technology, chapter Corporate Financialization., Oxford Wiley, 2015.
- Aglietta, M. (2018). *Money: 5000 years of debt and power*. Odile Jacob.
- Alijani, S., & Karyotis, C. (2016). *Sustainability: Finance, economy, and society*. UK: Emerald Group Publishing.
- Ardalan, K. (2008). *On the role of paradigms in finance. Alternative voices in contemporary economics*. Ashgate.
- Baker, M. P., Bergstresser, D. B., Serafeim, G., & Wurgler, J. A. (2018). *Financing the Response to Climate Change: The Pricing and Ownership of U.S. Green Bonds*. MA, USA: National Bureau of Economic Research.

- Bank of England. (2014). Money creation in the modern economy. *Quarterly Bulletin*, 2014, Q1.
- Bardi, U., Falsini, S., & Perissi, I. (2019). Toward a general theory of societal collapse: A biophysical examination of Tainter's model of the diminishing returns of complexity. *Biophys Econ Resour Qual*, 4, 3. <https://doi.org/10.1007/s41247-018-0049-0>
- Brière, M., Pouget, S., & Ureche-Rangau, L. (2019). Do Universal Owners Vote to Curb Costs of Asset Pricing Anomalies. SSRN Working Paper N° 3380239.
- Burrell, G., & Morgan, G. (1979). *Sociological paradigms and organizational analysis*. Aldershot: Gower Publishing Company Limited.
- Chambost, I., Lenglet, M., & Tadjeddine, Y. (2019). (Eds.) *the making of finance: Perspectives from the social sciences*. Oxon, UK: Routledge.
- Clapp, J., & Fuchs, D. A. (Eds.). (2009). *Corporate power in global agrifood governance*. MIT Press.
- Clark, W. C., & Munn, R. E. (1986). *Sustainable development of the biosphere*. UK: Cambridge university press (491 pages).
- Dafermos, M., & Nikolaidi, G. (2018). Galanis. Climate change, financial stability and monetary policy. *Ecological Economics*, 152, 219–234.
- Descola, P. (2005). *Par-delà nature et culture*. Gallimard, Paris: NRF.
- Dron, D. (2015). *Pour une régulation écosystémique de la finance*. Veblen Institute. https://www.veblen-institute.org/IMG/pdf/pour_une_regulation_ecosystemique.pdf (Accessed date: 9 May 2019).
- Earth, F. (2020). Our future on Earth. Science insight into our planet and society. Available at: <https://futureearth.org/publications/our-future-on-earth/>.
- Epstein, G. (2005). Introduction: Financialization and the world economy. *Financialization*, 3–16.
- Fama, E. F. (1991). Efficient capital markets: II. *The Journal of Finance*, 46, 1575–1617. <https://doi.org/10.1111/j.1540-6261.1991.tb04636.x>
- Folke, C., Biggs, R., Norström, A. V., Reyers, B., & Rockström, J. (2016). Social-ecological resilience and biosphere-based sustainability science. *Ecology and Society*, 21(3), 41. <https://doi.org/10.5751/ES-08748-210341>.
- Folke, C., et al. (2010). Resilience thinking: Integrating resilience, adaptability and transformativity. *Ecology and Society*, 15, 20–28.
- Friede, G., Busch, T., & Bassen, A. (2015). ESG and financial performance: aggregated evidence from more than 2000 empirical studies. *Journal of Sustainable Finance & Investment*, 5(4), 210–233.
- Friedman, M. (1953). The methodology of positive economics. In *Essays in Positive Economics* (pp. 153–184). Chicago, IL: Chicago University Press.
- Galaz, V., Gars, J., Moberg, F., Nykvist, B., & Repinski, C. (2015). Why ecologists should care about financial markets. *Trends in Ecology & Evolution*, 30, 571–580.
- Galaz, V., et al. (2018). Tax havens and global environmental degradation. *Nat. Ecol. Evol.*, 2, 1352–1357.
- Gimet, C., & Lagoarde-Segot, T. (September 2019). Reyes-Ortiz, L. 2019. Financialization and the macroeconomy. Theory and empirical evidence. *Economic Modelling*, 81, 89–110.
- Godley, W., & Lavoie, M. (2012). Monetary economics. In *An integrated approach to credit, money, income, production and wealth*. Palgrave-MacMillan, UK.
- Graeber, D. (2011). *Debt: The first 5,000 years*. Brooklyn, N.Y: Melville House (Print).
- Gurley, J. G., & Shaw, E. S. (1955). Financial aspects of economic development. *American Economic Review*, 45, 515–538.
- Jefferson, O. A., Köllhofer, D., Ehrlich, T. H., & Jefferson, R. A. (2015). The ownership question of plant gene and genome intellectual properties. *Nature Biotechnology*, 33, 1138–1143.
- Karpf, A., & Mandel, A. (2017). “Does it pay to be green?” working paper. Paris School of Economics.
- Koh, L. P., & Wilcove, D. S. (2008). Is oil palm agriculture really destroying tropical biodiversity? *Conservation Letters*, 1, 60–64.
- Krippner, G. R. (2005). The financialization of the American economy. *Socio-Economic Review*, 3(2), 173–208. <https://doi.org/10.1093/SER/mwi008>
- Kuhn, T. (1962). *The structure of scientific revolutions*. University of Chicago Press.
- Lagoarde-Segot, T. (2015). Diversifying financial research: From financialization to sustainability. *International Review of Financial Analysis*, 39, 1–6.
- Lagoarde-Segot, T. (2019). Sustainable finance. A critical realist perspective. *Research in International Business and Finance*, 47, 1–9.
- Lagoarde-Segot, T. (2020). Financing the sustainable development goals. *Sustainability*, 12, 2275.
- Lagoarde-Segot, T., & Currie, W. L. (2018). Financialization and information technology: A multi-paradigmatic view of IT and finance – Part II. *J Inf Technol*, 33, 1–8. <https://doi.org/10.1057/s41265-017-0045-7>
- Lagoarde-Segot, T., & Paraque, B. (2018). Finance and sustainability: from ideology to utopia. *International Review of Financial Analysis*, 55, 80–92.
- Lavoie, M. (2015). *Post-Keynesian economics*. Edward Elgar, USA: New Foundations.
- Lenton, T. M., & Latour, B. (2018). Gaia 2.0. *Science*, 361, 1066–1068.
- MacKenzie, D. (2006). *An engine, not a camera: How financial models shape markets*. Cambridge (Massachusetts): The MIT Press.
- Maitland, R. (2019). *Exploring emergence in corporate sustainability*. PhD Thesis. South Africa: University of Cape Town.
- Martínez, E. A. (2018). Aportes de la biología a la reflexión sobre modelos económicos alternativos: más allá de las metáforas. *Revista Estudios Nueva Economía*, 5(1), 55–69. In spanish, revised on line at: <http://estudiosnuevaeconomia.cl/team-view/estudios-nueva-economia-vii/>.
- Martínez, E. A., & Olguín, P. (2013). Transparency from production to consumption: New challenges for the quinoa market chain. In A. Bhargava, & S. Srivastava (Eds.), *Quinoa: Botany, production and uses* (pp. 234–239). Oxfordshire, UK: CABI Publishing, CAB International.
- Martínez, E. A., & Santelices, B. (1992). Size hierarchy and the –3/2 “Power law” relationship in a coalescing seaweed. *Journal of Phycology*, 28, 259–264.
- Meiksins-Wood, E. (2009). *L'origine du capitalisme*. Québec: Lux/Humanités.
- Meiksins-Wood, E. (2013). *Des citoyens aux seigneurs*. Lux/Humanités, Québec: Une histoire sociale de la pensée politique de l'antiquité au Moyen Âge.
- Meiksins-Wood, E. (2014). *Liberté et propriété. Une histoire sociale de la pensée politique occidentale de la Renaissance aux Lumières*. Québec: Lux/Humanités.
- Michie, J., & Oughton, C. (April 2013). Measuring diversity in financial services market: a diversity index. *Centre for Finance and Management Studies. DP Series. DP*, 113.
- Morgan, G. (1980). Paradigm, metaphors, and puzzle solving in organization theory. *Administrative Science Quarterly*, 25, 605–622.
- Nicholls, J., Lawlor, E., Neitzer, E., & Goodspeed, T. (2009). *A guide to social return on investment*.
- Nyström, J.-B., Jouffray, A., Norström, V., Crona, B., Søgaard-Jørgensen, P., Carpenter, S. R., ... Folke, C. (2019). Anatomy and resilience of the global production ecosystem. *Nature*, 575. <https://doi.org/10.1038/s41586-019-1712-3>
- Österblom, H., et al. (2015). Transnational corporations as “keystone actors” in marine ecosystems. *PLoS One*, 10, Article e0127533.
- Ostrom, E. (2005). *Understanding institutional diversity*. Princeton: Princeton University Press.
- Paraque, B., & Pérez, R. (Eds.). (2016). *Series: Critical Studies on Corporate Responsibility, Governance and Sustainability: 10. Finance reconsidered: New perspectives for a responsible and sustainable finance*. Emerald Group Publishing Limited.
- Rambaud, A., & Richard, J. (2015). The “triple depreciation line” instead of the “triple bottom line”: Towards a genuine integrated reporting. In , 33. *Critical Perspectives On Accounting, Special Issue on Social and Environmental Accounting* (pp. 92–116).
- Revelli, C. (2016). Re-embedding financial stakes within ethical and social values in socially responsible investing (SRI). *Research in International Business and Finance*, 38, 1–5.
- Rockström, J., Steffen, J. W., Noone, K., et al. (2009). Planetary boundaries: Exploring the safe operating space for humanity. *Ecology and Society*, 14, 32.
- Rodrigues, J. L. M., et al. (2013). Conversion of the Amazon rainforest to agriculture results in biotic homogenization of soil bacterial communities. *Proc. Natl Acad. Sci. USA*, 110, 988–993.
- Russi, L. (2013). *Hungry capital: The financialization of food*. Winchester, UK: Zero Books.
- SDSN. (2018). Closing the SDG budget gap. *Move Humanity*, (December 2018).
- Servet, J.-M. (2012). *Les monnaies du lien*. Presses Universitaires de Lyon.
- Simon, H. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society*, 106(6), 467–482. ISSN 0003-049X.
- Steffen, W., Persson, A., Deutsch, L., Zalasiewicz, I., Williams, M., Richardson, K., & Crumley, C. (2011). The Anthropocene: From global change to planetary stewardship. *Ambio: A Journal of the Human Environment*, 40(7), 739–761.
- Steffen, W., Sanderson, A., Tyson, P., Jäger, J., Matson, P. A., Moore, B., III, & Wasson, R. (2005). *Global change and the earth system. A planet under pressure*. Berlin: Springer.
- Stern, N. (2007). *The economics of climate change*. Cambridge, UK: The Stern review.
- Testart, A. (2007). *Critique du don, études sur la circulation non marchande*. Paris: Syllepse.
- Ulanowicz, R., Goerner, S. J., Lietaer, B., & Gomez, R. (2009). Quantifying sustainability: Resilience, efficiency and the return of information theory. *Ecological Complexity*, 6(1), 27–36.
- Ulanowicz, R. E. (2009). Increasing entropy: heat death or perpetual harmonies? *Int. J. of Design & Nature and Ecodynamics*, 4(2), 83–96.
- United Nations Statistical Commission. (2020). Report. available at: <https://unstats.un.org/unsd/statcom/51st-session/>.
- Vitali, S., Glattfelder, J. B., & Battiston, S. (2011). The network of global corporate control. *PLoS One*, 6(10), e25995.
- Walter, C. (2016). The financial logos: The framing of financial decision-making by mathematical modelling. *Research in International Business and Finance*, 37, 597–604.
- Walter, C. (2019). The leptokurtic crisis and the discontinuous turn in financial modelling. In I. Chambost, M. Lenglet, & Y. Tadjeddine (Eds.), *The making of finance. Perspectives from the social sciences*. Routledge.
- Yates, B. T., & Marra, M. (2017). Introduction: Social return on investment (SROI). *Evaluation and Program Planning*, 64, 95–97. <https://doi.org/10.1016/j.evalprogplan.2016.10.013>
- Zerbib, O. D. (2020). A sustainable capital asset pricing model (S-CAPM): Evidence from green investing and sin stock exclusion. Available at SSRN <https://ssrn.com/abstract=3455090>.